

命令行的一些知识

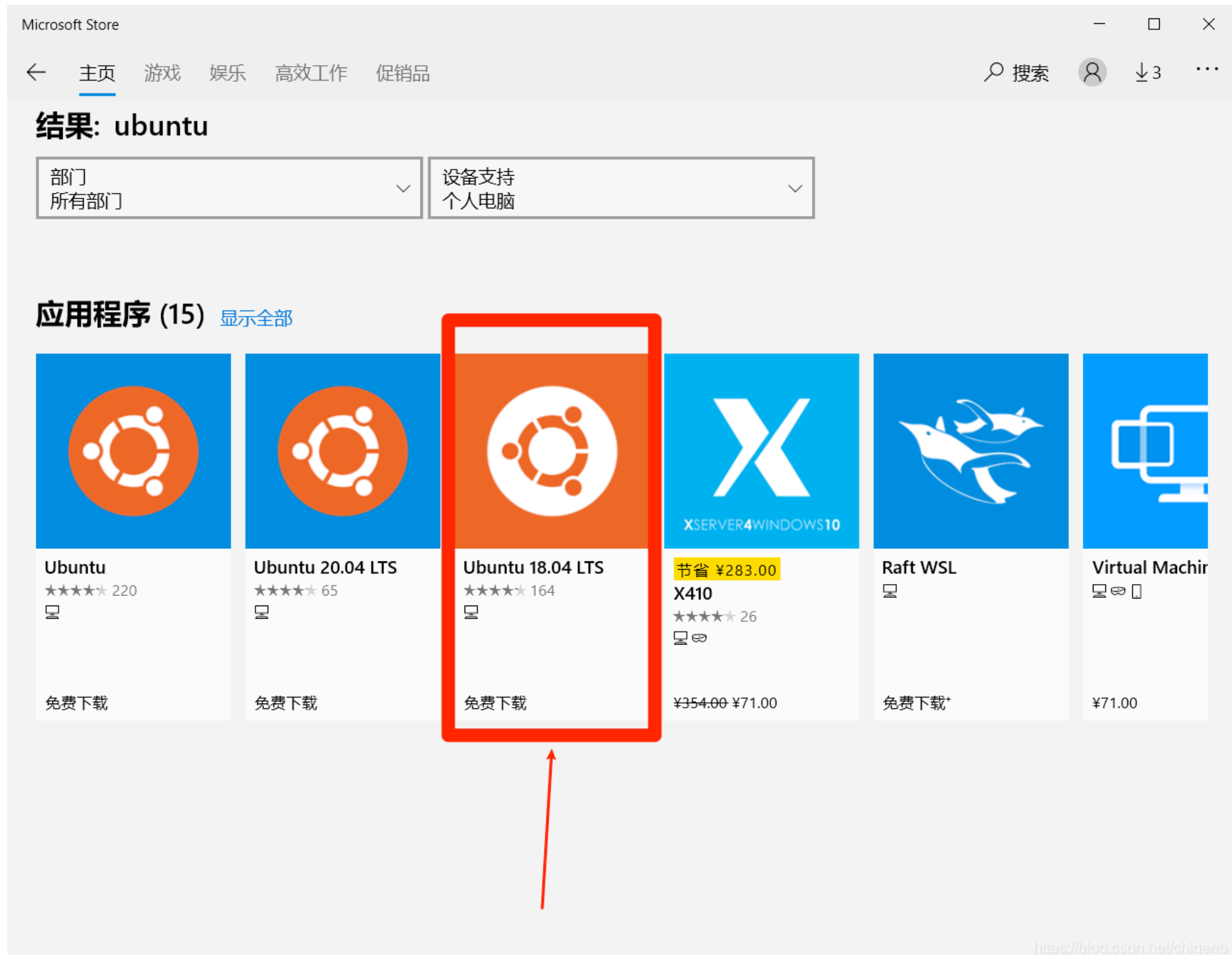


先把这个命令行的理念搞搞懂

Windows Subsystem for Linux (WSL)

- 1. 先进入控制面板->程序->启用或关闭Windows功能->打开“适用于Linux的Windows子系统”选项->点确定，然后电脑会让重启，重启就好。
- 2. 打开win10的Microsoft Store，搜索Ubuntu，下最新的 LTS （长期支持） 版本

Windows Subsystem for Linux (WSL)



Windows Subsystem for Linux (WSL)



设置一下，就会得到一个ubuntu的shell

```
zhang@LAPTOP-6UK545GK: ~  
Installing, this may take a few minutes...  
Please create a default UNIX user account. The username does not need to match your Windows username.  
For more information visit: https://aka.ms/wslusers  
Enter new UNIX username: zhang  ← 输入用户名，不能有大写  
Enter new UNIX password:      ← 密码看不见是正常的  
Retype new UNIX password:  
passwd: password updated successfully  
Installation successful!  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
zhang@LAPTOP-6UK545GK:~$ 安装完成
```

<https://blog.csdn.net/chigenb>

Missing Semester

日程

文档同步时间 2021-04-24

- 1/13: [课程概览与 shell](#)
- 1/14: [Shell 工具和脚本](#)
- 1/15: [编辑器 \(Vim\)](#)
- 1/16: [数据整理](#)
- 1/21: [命令行环境](#)
- 1/22: [版本控制\(Git\)](#)
- 1/23: [调试及性能分析](#)
- 1/27: [元编程](#)
- 1/28: [安全和密码学](#)
- 1/29: [大杂烩](#)
- 1/30: [提问&回答](#)

Solution	✓	Update	✓	Chinese	✓
Solution	✓	Update	✓	Chinese	✓
Solution	✓	Update	✓	Chinese	✓
Solution	✓	Update	✓	Chinese	✓
Solution	✓	Update	✓	Chinese	✓
Solution	✓	Update	✓	Chinese	✓
Solution	✓	Update	✓	Chinese	✓
Solution	✓	Update	✓	Chinese	✓
Solution	✗	Update	✓	Chinese	✓
Solution	✗	Update	✓	Chinese	✓

讲座视频可以在 [YouTube](#) 上找到。

<https://missing-semester-cn.github.io/>

<https://missing.csail.mit.edu/>

Shell

- 打开shell，就会得到类似如下的提示符：

```
example:~$
```

你现在工作在叫 example 的机器，以及当前你所在目录是 ~ (short for “home”).

\$ 说明你不是管理员 (否则是 #). 你可以通过敲击 “su” 来进入管理员。或者使用 “sudo XXX” 来使用管理员身份来执行命令 “XXX”.

Shell

输入一个命令，你就得到shell给你解释的结果

```
example:~$ date  
Feb 22 Jan 2022 11:49:31 AM EST  
example:~$
```


在shell下导航

```
example:~$ pwd
/home/example
example:~$ cd /home
example:/home$ pwd
/home
example:/home$ cd ..
example:/$ pwd
/
example:/$ cd ./home
example:/home$ pwd
/home
example:/home$ cd example
example:~$ pwd
/home/example
example:~$ ../../bin/echo hello
hello
```

看看当前目录下有哪些文件或目录

```
example:~$ ls
example:~$ cd ..
example:/home$ ls
example
example:/home$ cd ..
example:/$ ls
bin
boot
dev
etc
home
...
```

可以带一些参数，能够看到更多信息

```
example:~$ ls -l /home
drwxr-xr-x 1 example users 4096 Jun 15 2019 example
```

想知道某个命令怎么用

当然，也可以问baidu、
google、bing、
ChatGpt、
Stack overflow

```
example:~$ ls --help
```

```
usage: ls [-@ABCFGHILOPRSTUWabcdefghijklmnopqrstuvwxyz1%,] [--color=when] [-D format] [file ...]
```

```
example:~$ man ls
```

```
LS(1)                                General Commands Manual                                L

NAME
  ls - list directory contents

SYNOPSIS
  ls [-@ABCFGHILOPRSTUWabcdefghijklmnopqrstuvwxyz1%,] [--color=when] [-D format] [file ...]

DESCRIPTION
  For each operand that names a file of a type other than directory, ls displays its name as well as any requested, associated information. For each operand that names a file of type directory, ls displays the names of files contained within that directory, as well as any requested, associated information.

  If no operands are given, the contents of the current directory are displayed. If more than one operand is given, non-directory operands are displayed first; directory and non-directory operands are sorted separately and in lexicographical order.

  The following options are available:

  -@      Display extended attribute keys and sizes in long (-l) output.

  -A      Include directory entries whose names begin with a dot ('.') except for . and ..
          Automatically set for the super-user unless -I is specified.
```

Shell还可以重定向

```
example:~$ echo hello > hello.txt
example:~$ cat hello.txt
hello
example:~$ cat < hello.txt
hello
example:~$ cat < hello.txt > hello2.txt
example:~$ cat hello2.txt
hello
```